

Perform PostgreSQL VACUUM with VERBOSE and ANALYZE options

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Date: 2026-04-25

Executes PostgreSQL VACUUM operation with VERBOSE output and ANALYZE for statistics update, following best practices for database maintenance. Accepts arguments: username password database host.

Language

postgresql

Category

database

Command

```
PGPASSWORD="$2" psql -U "$1" -d "$3" -h "$4" -c "VACUUM (
    VERBOSE , ANALYZE) ;"
```

Explanation

The command connects to PostgreSQL via psql and runs VACUUM with VERBOSE (detailed output) and ANALYZE (update statistics). VACUUM reclaims storage occupied by dead tuples, while ANALYZE updates planner statistics. This combination is recommended for routine maintenance. Arguments: \$1=username, \$2=password (sets PGPASSWORD), \$3=database, \$4=host.

Tags

postgresql, database, maintenance, vacuum, optimization

Dependencies

postgresql-client

Arguments

1. **USERNAME** (Required): PostgreSQL username
2. **PASSWORD** (Required): PostgreSQL password (sets PGPASSWORD environment variable)
3. **DATABASE** (Required): Database name
4. **HOST** (Optional): Database host
Default: localhost

Examples

1. `vacuum_pg() { PGPASSWORD="$2" psql -U "$1" -d "$3" -h "$4" -c "VACUUM (VERBOSE, ANALYZE);"; }` - Define shell function for alias usage
2. `vacuum_pg postgres secret123 mydb localhost` - Call function with arguments: user password db host
3. `PGPASSWORD="secret456" psql -U "appuser" -d "production" -h "db.example.com" -c "VACUUM (VERBOSE, ANALYZE);"` - Direct command with explicit values

Output

```
VACUUM
INFO:  vacuuming "public.some_table"
INFO:  "some_table": removed 1000 row versions in 5 pages
INFO:  "some_table": found 1000 removable, 5000 nonremovable
        row versions
DETAIL:  0 dead row versions cannot be removed yet.
CPU 0.00s/0.00u sec elapsed 0.00 sec.
INFO:  analyzing "public.some_table"
INFO:  "some_table": scanned 3000 of 3000 pages, containing
        5000 live rows and 0 dead rows
VACUUM
```

Notes

- Run during periods of low database activity
- VERBOSE provides detailed output for monitoring
- ANALYZE updates statistics for query planner
- Consider using .pgpass file for password management

- Monitor `pg_stat_activity` for concurrent sessions
- For alias usage: `vacuum.pg() { PGPASSWORD="$2" psql -U "$1" -d "$3" -h "$4" -c "VACUUM (VERBOSE, ANALYZE);"; }`

Warnings

- VACUUM requires ACCESS SHARE lock on tables (conflicts with DDL)
- Avoid running during peak hours or critical operations
- For aggressive cleanup, consider VACUUM FULL but it requires exclusive lock
- Ensure adequate `maintenance_work_mem` setting for performance

See Also

- `postgresql-reindex`
- `postgresql-analyze-only`

Status

draft

Safety

caution

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

Created At

2026-04-25

Updated At

2026-04-25